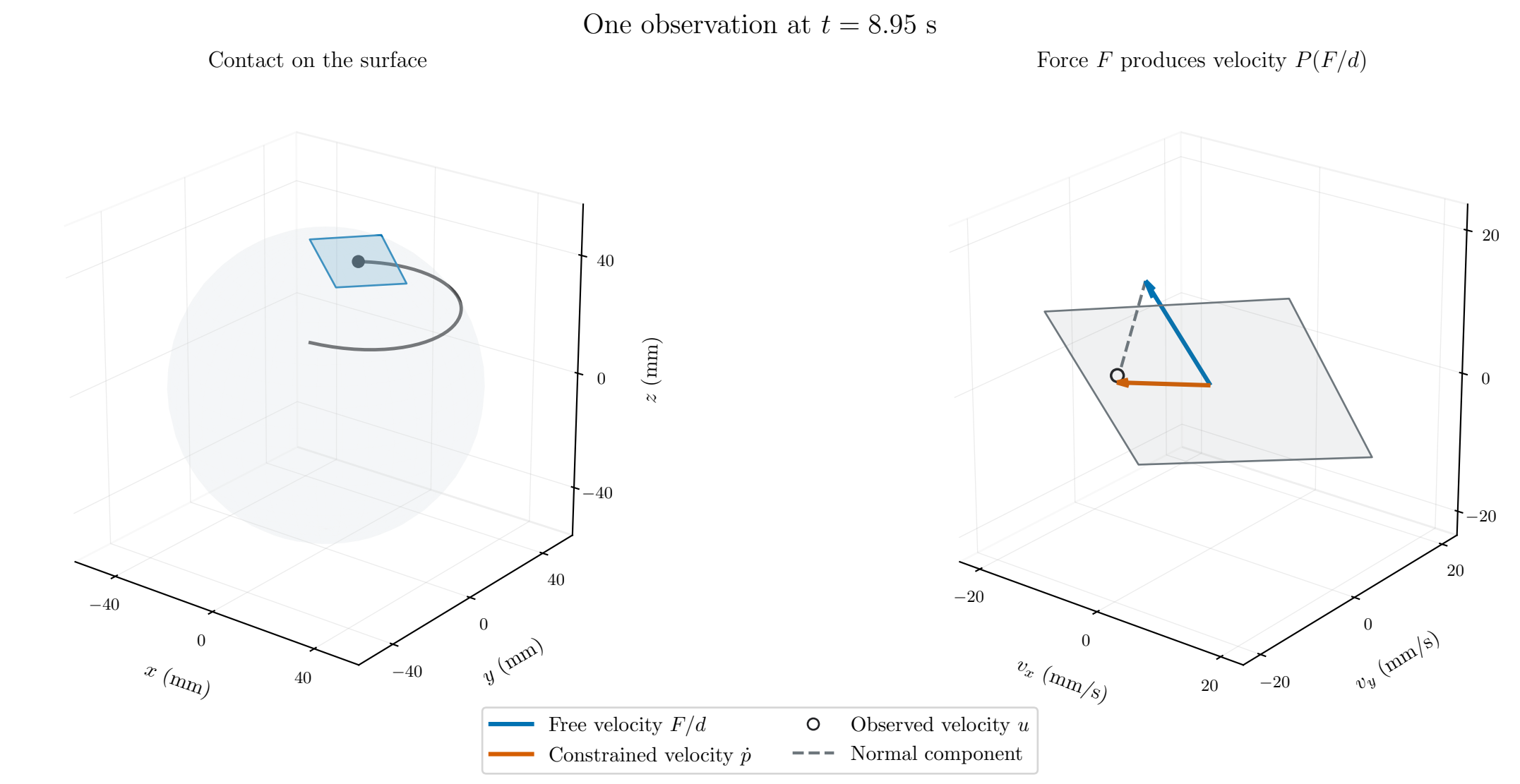

One observation at $t = 8.95$ s
Contact on the surface
Force $F$ produces velocity $P(F/d)$
40
0
−40
$z$ (mm)
−40
0
40
$x$ (mm)
40
0
−40
$y$ (mm)
20
0
−20
$v_z$ (mm/s)
−20
0
20
$v_x$ (mm/s)
20
0
−20
$v_y$ (mm/s)
Free velocity $F/d$
Constrained velocity $\dot{p}$
Observed velocity $u$
Normal component